A New Design for CUP to Reduce Reflection and Diffraction Based on LTPO and CFOT

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Abstract
Owing to their power-saving advantages, LTPO and CFOT technologies are increasingly adopted in smart phones, especially in large size foldable mobile phones. However, there are still some issues with the application of (Camera Under Panel) CUP technology in LTPO and CFOT mobile phones, such as high reflectivity and the balance between camera performance and display performance. Here, we present a new design for CUP technology applied in a 6.51-inch AMOLED panel. The reflectivity of the camera area is reduced to 7.5%, the difference in reflectivity between the camera area and the normal area is 1.8%, and the transmittance reaches 19.5%. The CUP PPI is 282, and all performances are quite good.

Author Keywords
Camera Under Panel (CUP); LTPO OLED; Color Filter On TPOT (CFOT); Low reflectivity; High transmittance

1. Introduction and Motivations
LTPO and CFOT technologies have become the configurations pursued by flagship smartphones due to their ability to reduce power consumption. This is particularly crucial for large foldable screens, where the increased display area demands even greater power saving. Consequently, most foldable screens adopt a combined LTPO+CFOT technology solution. On the other hand, full-screen displays remain a key pursuit for consumer electronics manufacturers. While products utilizing CUP technology are now available on the market, most still rely on POL solutions [1] [2]. Examples include Nubia Z70 Ultra and Lenovo's YOGA Air X AI Edition, both of which are POL products. This situation arises partly because CFOT technology has not yet achieved widespread adoption, and partly due to challenges in combining CFOT with CUP technology. Currently, among numerous CFOT products, only Samsung's Galaxy Fold has adopted CUP technology, though even Samsung removed this design in its latest Fold 7. Challenges in combining CFOT with CUP primarily include ring-shaped mura, high reflectivity, and poor display quality and photography performance. In this work, we took a series of measures to reduce reflectivity and enhance both photography and display quality.

2. Optimization of ring-shaped mura
We adopt a design where the CUP driving circuit is compressed and placed externally, using ITO transparent traces to connect the pixel circuit and the OLED device. Besides, to improve transmittance and maintain consistency with the L-decay between camera area and normal area, only a circle of BM is retained around the pixels in the camera area, while the BM in the rest of camera area is hollowed out.

After tuning the gamma of the CUP area, we found that there was still an obvious ring-shaped mura in the CUP area. As shown in Figure 1a. The phenomenon manifested as follows: the brightness of the inner circle was close to that of the Normal area, while the brightness of the outer circle was relatively dark. Meanwhile, the dark area corresponded to the CUP driving area, and the inner circle corresponded to the camera area. After further analysis, we discovered that it was the difference in BM area ratio between the driving area and the camera area that led to the difference in CF thickness between the two areas, which in turn resulted in the brightness difference. As shown in Figure 1b and 1c. The BM area ratio in the camera area was small, and the gaps between the BM of pixels needed to be filled during CF coating, which caused the CF above the pixels in this area to be relatively thin.

Before coating the CF, we add a layer of OC to fill the gaps in the BM. As shown in Figure 1d. Within the same area size, the volume of this OC plus the volume of the BM in the camera area is equal to the volume of the BM in the driving area. In this way, during the subsequent CF coating process, the consistency of the CF thickness in the two areas can be ensured, thereby eliminating the ring-shaped mura.

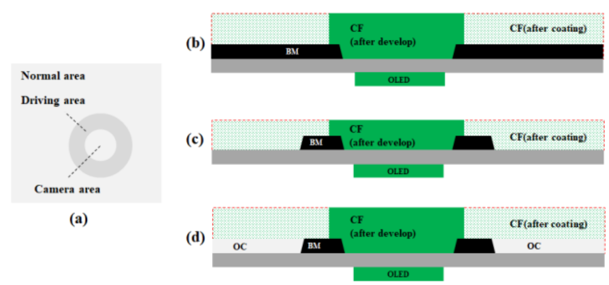

Figure 1. (a) The morphology of ring-shaped mura (b) driving area (c) camera area without OC (d) camera area filled with OC

3. Reduction of reflectivity
To improve transmittance, we increase the area of the light-transmitting region by means such as reducing the aperture ratio of the PDL, narrowing the width of the annular BM, and decreasing the distance between the BM and the edge of the PDL. Under this condition, the transmittance at 550 nm can reach 32.5%, but the reflectivity is as high as 21%, which seriously affects the visual effect. Therefore, we adopt the CPM (Cathode Patterning Material) scheme to remove the inter-pixel cathode [3], leaving only the part of the cathode that connects the pixels. This reduces the reflectivity to 13.7% and increases the transmittance to 40%. However, the reflectivity of the Normal area is 5.7%, and there is still a significant difference in reflectivity between the Camera area and the Normal area. As a result, the CUP area is still relatively noticeable to the naked eye.

The main reason for the high reflectivity is that the ambient light is reflected after irradiating the panel. In this work, we use gray OC instead of the aforementioned OC. The dye in Gray OC can absorb incident ambient light, reduce reflected light (Figure 2), and further lower the reflectivity of the camera area to 7.5% (Table 1). The visual effect has been significantly improved. In addition, although Gray OC causes a certain loss of transmittance, we have achieved a transmittance of 19.5% by

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adopting methods such as removing the inorganic layer between the double-layer PI, which can basically meet the needs of photography.

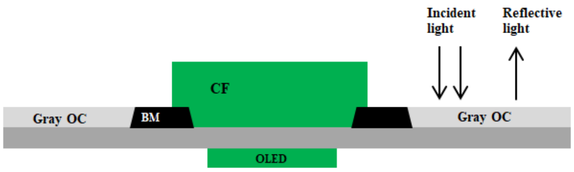

Figure 2. Explanation of Reflectivity Reduction by Gray OC

Table 1. The transmittance and reflectivity of different design

Camera area	Full cathode+OC	CPM+OC	CPM+Gray OC
Transmittance @550nm	32.5%	40%	19.5%
Reflectivity	21%	13.7%	7.5%

4. Optimization of pixel arrangement

To reduce diffraction effects, the CUP area adopts a pixel halving design. Initially, our design was to remove the RGBG subpixels within a 2×2 pixel region, retain the BGRG subpixels with uniform distribution, and use this 2×2 pixel unit as the minimum repeating unit of the CUP area (Figure 3a). We conducted white display simulations using drawing tools and found that this pixel arrangement exhibited color shift at the boundary of the normal area and the CUP area (Figure 3b). This is because the pixels at the CUP boundary only contain one or two colors, making it impossible to synthesize white light.

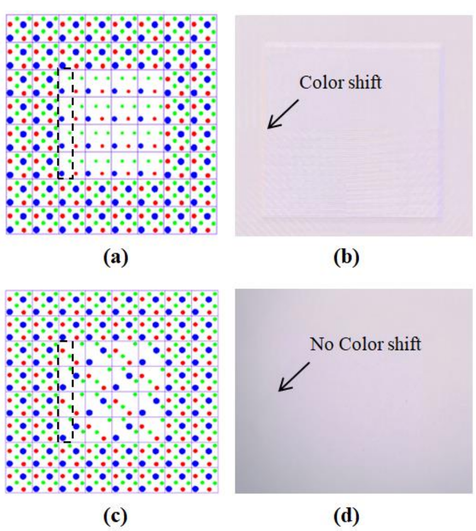

Figure 3. (a) The initial pixel arrangement and (b) corresponding simulation result, (c) The final pixel arrangement and (d)corresponding simulation result

Therefore, we proposed an improved design: within a 2×2 pixel region, remove either the BBGG or RRGG subpixels while retaining the corresponding RRGG or BBGG subpixels (with the positions of the retained subpixels unchanged), and adopt a 4×4 pixel unit as the minimum repeating unit of the CUP area (Figure 3c). With this improvement, all three RGB colors exsit at the CUP boundary, enabling white light synthesis. Similarly, we performed white display simulations using drawing tools and confirmed that this design eliminates color shift (Figure 3d). Consequently, it has been finalized as the ultimate pixel arrangement design.

5. Improvement of luminance uniformity

As shown in Figure 4, the OLED anodes in CUP region are connected to the driving circuits via ITO traces. Due to the varying distances between the driving circuits and their corresponding light-emitting devices, the lengths of the ITO traces also differ. This leads to variations in the RC characteristics of the pixel anodes, which in turn causes the issue of non-uniform pixel luminance.

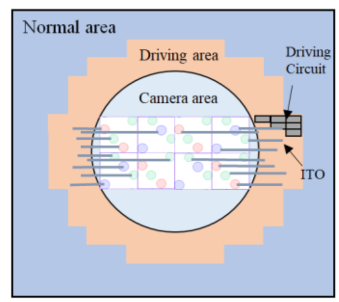

Figure 4. Layout schematic diagram of CUP area

To mitigate this problem, an anode load distribution compensation technique is adopted in this design. Figure 5(a) shows the layout of the ITO anode traces without compensation, while Figure 5(b) presents the optimized layout after compensation. Within the limited space, the ITO traces are arranged in a serpentine pattern to minimize the length difference between the longest and shortest paths.

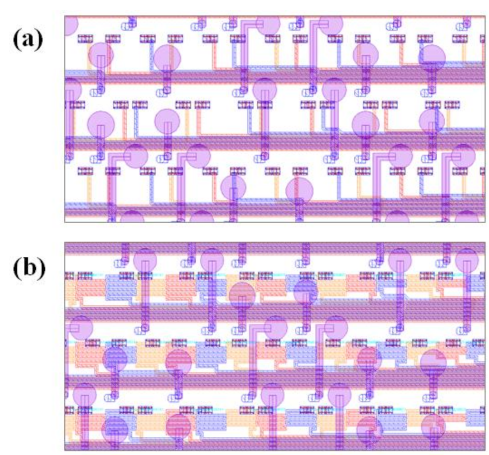

Figure 5. The layout of the ITO traces for the CUP area: (a) uncompendstated, (b) Compensated

The luminance uniformity has been experimentally verified through a dedicated uniformity test. As summarized in Table 2, a comparative analysis of the ITO trace lengths and luminance uniformity before and after compensation indicates that the luminance uniformity of 10 Gray in the CUP area is significantly improved, increasing from 75% to 82%.

Table 2. The length of ITO trace and the uniformity of luminance in the CUP area

Camera area	Uncompensated	Compensated
The shortest ITO trace/um	86	300
The medium ITO trace/um	890	1008
The longest ITO trace/um	1620	1620
The luminance uniformity of 10 Gray in CUP area	75%	82%

6. Shooting effect

We took photos using the normal mode of the front camera of Xiaomi Mi 11, and the shooting results are shown in Figure 6.

Figure 6. (a) Shooting with a bare lens (b) Shooting with our CUP panel

The images shot with our samples still exhibit favorable performance without diffraction algorithm enhancement [4] [5], with only exposure time compensation and color compensation were applied.

7. Conclusions

This work presents new design for CUP technology applied in LTPO and CFOT OLED display. The demo is shown in Figure 7, and the relevant characteristics are presented in Table 3. We eliminated the ring-shaped mura and reduced reflectivity by introducing Gray OC and CPM. Furthermore, the display performance was optimized through pixel arrangement and ITO traces compensation. This work provides a reference for the development of next-generation CUP technology.

Table 3. The characteristics of CUP panel

Item	Performance
Size	6.51 inch
TFT Backplane	LTPO+CFOT
Pixel circuit	7T2C
PPI(Normal/CUP)	399/282
Diameter(inner/outer)	3.0/6.1 mm
Transmittance @550nm	19.5%
Reflectivity(Normal/CUP)	5.7%/7.5%

Figure 7. (a) Display mode (b) Camera mode

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